

Dapagliflozin in Patients with Heart Failure and Reduced Ejection Fraction: The DAPA-HF Trial Findings

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ABSTRACT

BACKGROUND Sodium-glucose cotransporter 2 (SGLT2) inhibitors reduce the risk of heart failure hospitalization in patients with Type 2 Diabetes. We investigated whether Dapagliflozin would provide similar cardiovascular benefits in patients with established heart failure and a reduced ejection fraction, regardless of the presence of diabetes. **METHODS** In this phase 3, placebo-controlled trial, we randomly assigned 4744 patients with New York Heart Association class II, III, or IV heart failure and an ejection fraction of 40% or less to receive either Dapagliflozin 10 mg once daily or placebo, in addition to recommended therapy. The primary outcome was a composite of worsening heart failure (hospitalization or an urgent visit resulting in intravenous therapy for heart failure) or cardiovascular death. **RESULTS** Over a median of 18.2 months, the primary outcome occurred in 386 of 2373 patients (16.3%) in the Dapagliflozin group and in 502 of 2371 patients (21.2%) in the placebo group (hazard ratio, 0.74; 95% CI, 0.65 to 0.85; $P < 0.001$). A first worsening heart failure event occurred in 237 patients (10.0%) in the Dapagliflozin group and in 326 patients (13.7%) in the placebo group (HR 0.70; 95% CI, 0.59 to 0.83). **CONCLUSIONS** Among patients with heart failure and a reduced ejection fraction, the risk of worsening heart failure or death from cardiovascular causes was lower among those who received Dapagliflozin than among those who received placebo, regardless of the presence or absence of diabetes.

INTRODUCTION

SGLT2 inhibitors, such as Empagliflozin, Canagliflozin, and Dapagliflozin, were originally developed as glucose-lowering agents for Type 2 Diabetes. Large cardiovascular outcome trials (CVOTs), such as EMPA-REG OUTCOME and DECLARE-TIMI 58, surprisingly demonstrated that these agents robustly reduced the risk of hospitalization for heart failure.

This observation led to the hypothesis that SGLT2 inhibitors might benefit patients with established Heart Failure (HF) through mechanisms independent of glucose lowering. This trial (DAPA-HF) was designed to test the efficacy and safety of Dapagliflozin specifically in patients with chronic heart failure with reduced ejection fraction (HFrEF), including those without diabetes.

METHODS

STUDY POPULATION Eligible patients were 18 years of age or older, had New York Heart Association (NYHA) class II to IV heart failure, and had a left ventricular ejection fraction (LVEF) of $\leq 40\%$. Patients were required to have an elevated N-terminal pro-B-type natriuretic peptide (NT-proBNP) level (≥ 600 pg/ml). Key exclusion criteria included Type 1 Diabetes, systolic blood pressure < 95 mm Hg, and estimated glomerular filtration rate (eGFR) < 30 ml/min/1.73m².

PROCEDURES Patients were randomly assigned to receive Dapagliflozin 10 mg once daily or matching placebo. All patients continued their standard heart failure therapies (standard of care),

which included ACE inhibitors, ARBs, angiotensin receptor-neprilysin inhibitors (ARNI) like Sacubitril-Valsartan, beta-blockers, and mineralocorticoid receptor antagonists (MRAs).

STATISTICAL ANALYSIS The primary efficacy outcome was analyzed with use of a Cox proportional-hazards model stratified according to diabetes status at baseline. Kaplan-Meier estimates of the cumulative incidence of the primary outcome were calculated.

RESULTS

PRIMARY OUTCOME The primary composite outcome (worsening heart failure or cardiovascular death) occurred in 386 patients (16.3%) in the Dapagliflozin group and 502 patients (21.2%) in the placebo group (Hazard Ratio [HR] 0.74; 95% Confidence Interval [CI] 0.65-0.85; $P < 0.001$). The number needed to treat to prevent one primary outcome event was 21.

The benefit was consistent across prespecified subgroups, including patients with diabetes (HR 0.75) and without diabetes (HR 0.73), confirming that the mechanism of action is independent of glycemic control.

SECONDARY OUTCOMES Cardiovascular death occurred in 227 patients (9.6%) in the Dapagliflozin group and 273 (11.5%) in the placebo group (HR 0.82; 95% CI 0.69-0.98). Hospitalization for heart failure occurred in 231 (9.7%) vs 318 (13.4%) respectively (HR 0.70; 95% CI 0.59-0.83).

All-cause mortality was also lower in the Dapagliflozin group (276 deaths, 11.6%) than in the placebo group (329 deaths, 13.9%) (HR 0.83; 95% CI 0.71-0.97).

Kansas City Cardiomyopathy Questionnaire (KCCQ) scores, a measure of quality of life, improved significantly more in the Dapagliflozin group than in the placebo group.

DISCUSSION

These findings establish SGLT2 inhibitors as a new foundation or "pillar" of medical therapy for HFrEF, joining beta-blockers, RAAS inhibitors (ACEi/ARB/ARNI), and MRAs. The magnitude of benefit—a 26% reduction in the primary endpoint relative to excellent background therapy—is clinically substantial.

Mechanistically, the benefits of SGLT2 inhibitors in heart failure are complex and multifactorial. Proposed mechanisms include osmotic diuresis and natriuresis (reducing preload), reduction in arterial stiffness and vascular resistance (reducing afterload), improved myocardial energetics (shift towards ketone body utilization), and inhibition of the sodium-hydrogen exchanger in the myocardium.

The consistency of benefit in patients without diabetes is particularly important, as it expands the indication for these drugs to a much broader population of heart failure patients.

Similar benefits have been observed with Empagliflozin in the EMPEROR-Reduced trial, suggesting a class effect for SGLT2 inhibitors in HFrEF. However, Dapagliflozin was the first to demonstrate a significant reduction in cardiovascular death in this population.

CONCLUSION

Dapagliflozin reduced the risk of worsening heart failure and death from cardiovascular causes in patients with heart failure and reduced ejection fraction. These benefits were observed in patients receiving excellent standard therapy and extended to patients both with and without Type 2 Diabetes.

REFERENCES

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